LEFT AND RIGHT OBSERVATIONAL METRICS OF DIFFERENTIAL LOGICAL RELATIONS AND COMPOSITIONAL FUNCTION-DIFFERENCE BOUNDS

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Allowed standard prerequisites (editorial).

- Quantales, residuation, cartesian closed categories and simply typed lambda calculus. Full author §2–§4 definitions retained; cited background remains in complete source bibliography. Source-local metric and logical-relation constructions are retained as author material.

2 From Logical Relations to Differential Logical Relations

In this section we recall how differential logical relations can be seen as a quantitative generalization of standard logical relations, at the same time highlighting the metric counterparts of qualitative notions like equivalences and preorders. Moreover, we introduce quasi-quasi-metrics as the metric counterpart of quasi-reflexive and transitive relations. For simplicity, we identify lambda terms using the standard β-equalities throughout this section.

Logical Relations.

The theory of logical relations is well-known and has been exploited in various directions to establish qualitative properties of type systems, like e.g. termination [18], bisimulation [33] or parametricity [30, 21]. The idea is to start from some basic binary relation $\rho_o \subseteq o \times o$ over the terms of some ground type o. The relation ρ_o can then be lifted to a family of binary relations $\rho_{\mathsf{A}} \subseteq \mathsf{A} \times \mathsf{A}$, where A varies over all simple types constructed starting from o (indeed, one may consider recursive [14], polymorphic [32, 30] or monadic [20] types as well, but we here limit our discussion to simple types). The lifting is defined inductively by:

$(\mathsf{t}, \mathsf{t}') \in \rho_{\mathsf{A}\times\mathsf{B}} \iff (\mathsf{fst}(\mathsf{t}), \mathsf{fst}(\mathsf{t}')) \in \rho_{\mathsf{A}}$ and $(\mathsf{snd}(\mathsf{t}), \mathsf{snd}(\mathsf{t}')) \in \rho_{\mathsf{B}}$, $(\wedge)$ $(\mathsf{t}, \mathsf{t}') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}} \iff (\forall \mathsf{s}, \mathsf{s}' \in \mathsf{A})\ (\mathsf{s}, \mathsf{s}') \in \rho_{\mathsf{A}} \Rightarrow (\mathsf{t}\,\mathsf{s}, \mathsf{t}'\,\mathsf{s}') \in \rho_{\mathsf{B}}$. $(\Rightarrow)$ Typically, one wishes to establish a so-called fundamental lemma, stating that well-typed programs $x : \mathsf{A} \vdash \mathsf{t} : \mathsf{B}$ preserve relations. This means that, for any choice of a family of logical relations ρ_{A} defined as above, one can prove

$(\forall \mathsf{s}, \mathsf{s}' \in \mathsf{A})\ (\mathsf{s}, \mathsf{s}') \in \rho_{\mathsf{A}} \Rightarrow (\mathsf{t}[\mathsf{s}/\mathsf{x}], \mathsf{t}[\mathsf{s}'/\mathsf{x}]) \in \rho_{\mathsf{B}}$. (Fundamental Lemma) Notice that this is equivalent to the instance of reflexivity $(\lambda\mathsf{x}.\mathsf{t}, \lambda\mathsf{x}.\mathsf{t}) \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$.

Of particular interest are the equivalence relations (that is, those which are reflexive, symmetric and transitive) and the preorders (that is, the reflexive and transitive ones). We here focus on the latter, as we will not consider symmetry in this paper (see Remark 3). A fundamental observation is that the logical relation lifting preserves preorders (and indeed, equivalences): if ρ_{A} and ρ_{B} are reflexive and transitive, then $\rho_{\mathsf{A}\times\mathsf{B}}$ and $\rho_{\mathsf{A}\Rightarrow\mathsf{B}}$ can be shown reflexive and transitive as well. The case of the function space crucially relies on the fact that all programs of type $\mathsf{A} \Rightarrow \mathsf{B}$ must preserve relations (or, in other words, that the fundamental lemma holds): as

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we observed above, the reflexivity condition $(\mathsf{t},\mathsf{t}) \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$ coincides with the fact that the function t is relation-preserving; transitivity, instead, can be proved by combining relation-preservation, the reflexivity of ρ_{A} and the transitivity of ρ_{B}.

Any logical relation $\rho \subseteq \mathsf{A}\times\mathsf{A}$ induces an equivalence $\equiv_\rho$, called the observational equivalence, where $\mathsf{t} \equiv_\rho \mathsf{u}$ iff for all $\mathsf{s} \in \mathsf{A}$, $(\mathsf{s},\mathsf{t}) \in \rho$ iff $(\mathsf{s},\mathsf{u}) \in \rho$. Intuitively, two terms t,u are equivalent if the relation ρ cannot distinguish them. For example, if the definition of ρ_o on basic types only depends on the values $\mathsf{t} \Rightarrow^* \mathsf{v}$ produced by terms, one can usually deduce that terms are indistinguishable from their associated values, that is $\mathsf{t} \equiv_\rho \mathsf{v}$. In the absence of symmetry, one obtains two observational preorders $\sqsubseteq^l_\rho, \sqsubseteq^r_\rho \subseteq \mathsf{A}\times\mathsf{A}$ defined by:

$\mathsf{s} \sqsubseteq^l_\rho \mathsf{t} \iff (\forall \mathsf{u} \in \mathsf{A})\ (\mathsf{t},\mathsf{u}) \in \rho \Rightarrow (\mathsf{s},\mathsf{u}) \in \rho$, $\mathsf{s} \sqsubseteq^r_\rho \mathsf{t} \iff (\forall \mathsf{u} \in \mathsf{A})\ (\mathsf{u},\mathsf{s}) \in \rho \Rightarrow (\mathsf{u},\mathsf{t}) \in \rho$. These preorders satisfy the following useful and easily provable properties:

Proposition 1.

For any binary relation $\rho \subseteq \mathsf{A}\times\mathsf{A}$ and $c \in \{l, r\}$,

(i.)

$\sqsubseteq^c_\rho \supseteq \rho$ iff ρ is transitive;

(ii.)

$\sqsubseteq^c_\rho \subseteq \rho$ iff ρ is reflexive;

(iii.)

$\sqsubseteq^c_\rho = \rho$ iff ρ is a preorder;

(iv.)

The following hold:

$(\forall \mathsf{s},\mathsf{t},\mathsf{u} \in \mathsf{A})\ \mathsf{s} \sqsubseteq^l_\rho \mathsf{t} \wedge (\mathsf{t},\mathsf{u}) \in \rho \Rightarrow (\mathsf{s},\mathsf{u}) \in \rho$, (left transitivity) $(\forall \mathsf{s},\mathsf{t},\mathsf{u} \in \mathsf{A})\ (\mathsf{s},\mathsf{t}) \in \rho \wedge \mathsf{t} \sqsubseteq^r_\rho \mathsf{u} \Rightarrow (\mathsf{s},\mathsf{u}) \in \rho$. (right transitivity)

The reason why we delve into these basic properties of preorders is that we will soon explore their (less trivial!) quantitative counterparts, that arise naturally in the theory of differential logical relations. In particular, the left and right transitivity conditions will correspond to stronger variants of the triangular inequality for metric spaces.

Beyond preorders, we are interested in the following weaker notion:

Definition 2 (Quasi-Preorder).

A relation $\preceq \subseteq \mathsf{A}\times\mathsf{A}$ is called a quasi-preorder if it is transitive and (left-)quasi-reflexive, that is, $\mathsf{t} \preceq \mathsf{u} \Rightarrow \mathsf{t} \preceq \mathsf{t}$.

Quasi-preorders are obtained by weakening the reflexivity condition of preorders: intuitively, only the points which are smaller than someone are smaller than themselves. One can easily develop a theory of logical relations for quasi-preorders. The sole delicate point is that, in order to let such relations lift to function spaces, one has to slightly modify the relation lifting as follows:

$(\mathsf{t},\mathsf{t}') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}} \iff (\forall \mathsf{s},\mathsf{s}' \in \mathsf{A})\ (\mathsf{s},\mathsf{s}') \in \rho_{\mathsf{A}} \Rightarrow (\mathsf{t}\,\mathsf{s},\mathsf{t}'\,\mathsf{s}') \in \rho_{\mathsf{B}} \wedge (\mathsf{t}\,\mathsf{s},\mathsf{t}\,\mathsf{s}') \in \rho_{\mathsf{B}}$. ($\Rightarrow^{\mathrm{q}}$) Compared to ($\Rightarrow$ ▶ 2), ($\Rightarrow^{\mathrm{q}}$) includes a second clause $(\mathsf{t}\,\mathsf{s},\mathsf{t}\,\mathsf{s}') \in \rho_{\mathsf{B}}$ relating the action of t on both s and s'. With this definition, one can easily check that if $\rho_{\mathsf{A}}, \rho_{\mathsf{B}}$ are quasi-preorders, and the fundamental lemma holds, then $\rho_{\mathsf{A}\times\mathsf{B}}$ and $\rho_{\mathsf{A}\Rightarrow\mathsf{B}}$ are quasi-preorders as well.

Differential Logical Relations.

We now have all elements to discuss what happens when extending logical relations to a quantitative setting. Rather than considering binary relations $\rho \subseteq \mathsf{A}\times\mathsf{A}$ expressing that a certain property holds for two terms s,t or not, we will consider ternary relations $\rho \subseteq \mathsf{A}\times\mathcal{Q}_{\mathsf{A}}\times\mathsf{A}$, where $(\mathsf{s},a,\mathsf{t}) \in \rho$ indicates that a certain relation holds of s,t to a certain extent, quantified via $a \in \mathcal{Q}_{\mathsf{A}}$. Here $\mathcal{Q}_{\mathsf{A}}$ is a quantale, an

algebraic structure (recalled in the next section) that captures several properties of quantities as expressed by e.g. non-negative real numbers.

In fact, just like for standard logical relations, a differential logical relation $\rho_o \subseteq o \times \mathcal{Q}_o \times o$ on a ground type can be lifted to a family of binary relations $\rho_{\mathsf{A}} \subseteq \mathsf{A} \times \mathcal{Q}_{\mathsf{A}} \times \mathsf{A}$ over simple types. First, we define, by induction, the quantales $\mathcal{Q}_{\mathsf{A}\times\mathsf{B}} = \mathcal{Q}_{\mathsf{A}} \times \mathcal{Q}_{\mathsf{B}}$ and $\mathcal{Q}_{\mathsf{A}\Rightarrow\mathsf{B}} = \mathsf{A} \Rightarrow (\mathcal{Q}_{\mathsf{A}} \multimap \mathcal{Q}_{\mathsf{B}})$, where $\mathcal{Q}_{\mathsf{A}} \multimap \mathcal{Q}_{\mathsf{B}}$ is the quantale of monotone functions, and $\mathsf{A} \Rightarrow \mathcal{Q}_{\mathsf{A}} \multimap \mathcal{Q}_{\mathsf{B}}$ is the quantale of functions from the set of closed terms of type A to $\mathcal{Q}_{\mathsf{A}} \multimap \mathcal{Q}_{\mathsf{B}}$ equipped with the pointwise order. We then define the lifting of ρ_o by:

$((\mathsf{t},\mathsf{u}),(a,b),(\mathsf{t}',\mathsf{u}')) \in \rho_{\mathsf{A}\times\mathsf{B}} \iff (\mathsf{t},a,\mathsf{t}') \in \rho_{\mathsf{A}}$ and $(\mathsf{u},b,\mathsf{u}') \in \rho_{\mathsf{A}}$, $(\mathsf{t},f,\mathsf{t}') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}} \iff (\forall \mathsf{s},\mathsf{s}' \in \mathsf{A}, \forall a \in \mathcal{Q}_{\mathsf{A}})$ if $(\mathsf{s},a,\mathsf{s}') \in \rho_{\mathsf{A}}$, then $(\mathsf{t}\,\mathsf{s}, f(\mathsf{s})(a), \mathsf{t}\,\mathsf{s}') \in \rho_{\mathsf{B}}$ and $(\mathsf{t}\,\mathsf{s}, f(\mathsf{s})(a), \mathsf{t}'\,\mathsf{s}') \in \rho_{\mathsf{B}}$. Notice that the definition of $\rho_{\mathsf{A}\Rightarrow\mathsf{B}}$ closely imitates the clause $(\Rightarrow^{\mathrm{q}})$ for quasi-preorders. Also observe that the quantale $\mathcal{Q}_{\mathsf{A}\Rightarrow\mathsf{B}}$ for the function type is itself a set of functions relating terms of type A and quantities in $\mathcal{Q}_{\mathsf{A}}$ with quantities in $\mathcal{Q}_{\mathsf{B}}$. As we show in Section 5, this definition gives rise to an interpretation of the simply typed λ-calculus where a fundamental lemma holds under the following form: for all terms $x : \mathsf{A} \vdash \mathsf{t} : \mathsf{B}$ and choice of a family of differential logical relations ρ_{A} as above, there exists a map $\mathsf{t}^\bullet : \mathsf{A} \Rightarrow (\mathcal{Q}_{\mathsf{A}} \multimap \mathcal{Q}_{\mathsf{B}})$ such that

$$(\forall \mathsf{s},\mathsf{s}' \in \mathsf{A}, \forall a \in \mathcal{Q}_{\mathsf{A}})\ (\mathsf{s},a,\mathsf{s}') \in \rho_{\mathsf{A}} \ \Rightarrow\ (\mathsf{t}\,\mathsf{s}, \mathsf{t}^\bullet(\mathsf{s})(a), \mathsf{t}\,\mathsf{s}') \in \rho_{\mathsf{B}}.$$

(fundamental lemma) The function $\mathsf{t}^\bullet$ behaves like some sort of derivative of t: it relates errors in input with errors in output. This connection is investigated in more detail in [9, 29].

So far, everything works just as in the standard, qualitative, case. However, the quantitative setting is well visible when we consider the corresponding notions of equivalences and preorders. Recall that an (integral) quantale is, in particular, an ordered monoid $(\mathcal{Q}, +, 0, \leq)$ of which 0 is the minimum element. For a differential logical relation $\rho \subseteq \mathsf{A} \times \mathcal{Q}_{\mathsf{A}} \times \mathsf{A}$, reflexivity, symmetry and transitivity translate into the following conditions:

$(\forall \mathsf{t} \in \mathsf{A})\ (\mathsf{t},0,\mathsf{t}) \in \rho$, (reflexivity) $(\forall \mathsf{t},\mathsf{u} \in \mathsf{A}, \forall a \in \mathcal{Q}_{\mathsf{A}})\ (\mathsf{t},a,\mathsf{u}) \in \rho \ \Rightarrow\ (\mathsf{u},a,\mathsf{t}) \in \rho$, (symmetry) $(\forall \mathsf{s},\mathsf{t},\mathsf{u} \in \mathsf{A}, \forall a,b \in \mathcal{Q}_{\mathsf{A}})\ (\mathsf{s},a,\mathsf{t}) \in \rho \wedge (\mathsf{t},b,\mathsf{u}) \in \rho \ \Rightarrow\ (\mathsf{s},a{+}b,\mathsf{u}) \in \rho$. (transitivity) It is clear then that equivalence relations translate, in the quantitative setting, into some kind of metric space. Similarly, the quantitative counterpart of preorders are the so-called quasi-metric spaces [19], essentially, metrics without a symmetry condition, indeed a very well-studied class of metrics. In particular, we will show that, similarly to preorders, any ternary relation $\rho \subseteq \mathsf{A} \times \mathcal{Q}_{\mathsf{A}} \times \mathsf{A}$ gives rise to left and right observational quasi-metrics $\rho^l, \rho^r : \mathsf{A} \times \mathsf{A} \to \mathcal{Q}_{\mathsf{A}}$ satisfying properties analogous to those of Proposition 1.

► **Remark 3.**

While in the original definition [11] differential logical relations were symmetric, symmetry was abandoned in all subsequent works. The first reason is that several interesting notions of program difference, like e.g. those arising from incremental computing [9, 6, 1], are not symmetric. A second reason is that the cartesian closure is problematic in presence of both quasi-reflexivity and symmetry [29].

There is, however, an important point on which differential logical relations differ from standard logical relations: while the former lift preorders well to all simple types, their quantitative counterpart, the quasi-metrics, are not preserved by the higher-order lifting of differential logical relations. Indeed, we observed that an essential ingredient in the lifting of the reflexivity property is the fundamental lemma; yet, in the framework of differential logical relations, the fundamental lemma produces, for any term $\mathsf{t} : \mathsf{A} \Rightarrow \mathsf{B}$, the "reflexivity" condition $(\mathsf{t},\mathsf{t}^\bullet,\mathsf{t}) \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$, which

differs from standard reflexivity in that the distance is $\mathsf{t}^\bullet$ and not the minimum element 0. This means that the metric structure arising from differential logical relation cannot be that of standard (quasi-)metric spaces. Rather, it must be something close to the partial metric spaces [4, 23], that is, metric spaces in which the condition $d(x,x) = 0$ is replaced by the quasi-reflexivity condition $d(x,x) \leq d(x,y)$. We will discuss the connections with partial metric spaces in the next sections.

By replacing reflexivity with quasi-reflexivity, we obtain the quantitative counterpart of quasi-preorders, that we call qq-metrics (being "quasi" both in the sense of quasi-metrics, i.e. the rejection of symmetry, and of quasi-preorders, i.e. the weakening of reflexivity).

Definition 4.

For a set X and a quantale $\mathcal{Q}$, a relation $\rho \subseteq X \times \mathcal{Q} \times X$ is called quasi-quasi-metric (or more concisely qq-metric) if it is transitive and satisfies the condition

$(\forall x, y \in X, \forall a \in \mathcal{Q})\ (x,a,y) \in \rho \ \Rightarrow\ (x,a,x) \in \rho$. (quasi-reflexivity)

As shown in Section 4, the qq-metrics capture the properties of distances which are preserved by differential logical relations: indeed, the argument showing that the quasi-preorders lift to all simple types scales well to the quantitative setting, showing that a qq-metric on the base types gives rise to qq-metrics on all simple types.

The obvious question, however, is: what are these qq-metrics? How are they related to the more standard quasi-metrics and partial metrics? This is what we are going to do in the following section.

3 Quasi-Quasi-Metric Spaces

In this section we use the language of quantale-valued relations to explore the connections between the qq-metrics introduced in the previous section and the more well-established notions of quasi-metric and partial quasi-metric spaces.

Quantale-Valued Relations.

Let us recall that a quantale $\mathcal{Q}$ is a complete lattice $(\mathcal{Q}, \sqsubseteq)$ endowed with a continuous monoidal operation $\otimes$, with unit 1. Here, $\otimes$ being continuous means that both $x \otimes (-)$ and $(-) \otimes x$ preserve arbitrary suprema for all $x \in Q$. A quantale $\mathcal{Q}$ is integral (cf. [22], p. 148) when $1 = \top$ and commutative when $\otimes$ is commutative. Suppose $\mathcal{Q}$ is commutative. Given $x, y \in \mathcal{Q}$, their residual is defined as $x \multimap y := \bigvee\{z \in \mathcal{Q} \mid z \otimes x \sqsubseteq y\}$ where $\sqsubseteq$ is the partial order of $\mathcal{Q}$. Notice that $z \sqsubseteq x \multimap y$ iff $z \otimes x \sqsubseteq y$, and that $(x \multimap y) \otimes x \sqsubseteq y \sqsubseteq x \multimap (y \otimes x)$. A commutative quantale $\mathcal{Q}$ is divisible [23] if for all $x, y \in \mathcal{Q}$, $x \sqsubseteq y$ holds iff $y \otimes (y \multimap x) = x$. Equivalently, $\mathcal{Q}$ is divisible iff, whenever $x \sqsubseteq y$, there exists z such that $x = y \otimes z$. In the following we will use $\mathcal{Q}$ to refer to a commutative, integral and divisible quantale.

Example 5.

The Lawvere quantale is formed by the non-negative extended reals $[0, +\infty]$ with the reversed order $x \sqsubseteq y := x \geq y$, and with addition as monoidal operation. Notice that the ordering of quantales is reversed with respect to usual metric intuitions: the "0" element is the $\top$, joins correspond to taking infs, etc.

Given a quantale $\mathcal{Q}$ and sets X, Y, a $\mathcal{Q}$-relation over X, Y is a map $s\colon X \times Y \to \mathcal{Q}$, which can be visualized as a matrix with values in $\mathcal{Q}$. For $\mathcal{Q}$-relations $s, t\colon X \times Y \to \mathcal{Q}$, we write $s \sqsubseteq t$ when $s(x,y) \sqsubseteq t(x,y)$ for all $x \in X$ and $y \in Y$. Given $\mathcal{Q}$-relations $s\colon X \times Y \to \mathcal{Q}$, $t\colon Y \times Z \to \mathcal{Q}$ and $u\colon X \times Z \to \mathcal{Q}$, $w\colon Z \times Y \to \mathcal{Q}$, we define the $\mathcal{Q}$-relations $s \otimes t\colon X \times Z \to \mathcal{Q}$ and $u \multimap s\colon Z \times Y \to \mathcal{Q}$ and $s \mathrel{-\!\circ} w\colon X \times Z \to \mathcal{Q}$ via the two operations:

$(s \otimes t)(x,z) = \bigvee_{y \in Y} s(x,y) \otimes t(y,z)$, $(u \multimap s)(z,y) = \bigwedge_{x \in X} u(x,z) \multimap s(x,y)$, $(s \circ\!\!- w)(x,z) = \bigwedge_{y \in Y} w(z,y) \multimap s(x,y)$. The monoidal product $\otimes$ and the residuals $\multimap, \circ\!\!-$ of $\mathcal{Q}$-relations satisfy properties analogous to residuals in $\mathcal{Q}$, e.g. $s \otimes (s \multimap t) \sqsubseteq t$, $(t \circ\!\!- s) \otimes s \sqsubseteq t$. It is well-known that $\mathcal{Q}$-relations form a category $\mathcal{Q}\mathbf{Rel}$ whose objects are sets and such that $\mathcal{Q}\mathbf{Rel}(X,Y)$ are the $\mathcal{Q}$-relations from X to Y. The operation $s \otimes t$ is the composition operator of this category, while the identities are the relations defined as $\mathbf{1}_X(x,x) = 1 = \top$ and $\mathbf{1}_X(x, y \neq x) = \bot$.

Finally, for any relation $s \in \mathcal{Q}\mathbf{Rel}(X,X)$, define the relations $\Delta_1 s, \Delta_2 s \in \mathcal{Q}\mathbf{Rel}(X,X)$ by $\Delta_1 s = s \circ \Delta \circ \pi_1$ and $\Delta_2 s = s \circ \Delta \circ \pi_2$, that is, $\Delta_1 s(x,y) := s(x,x)$, $\Delta_2 s(x,y) = s(y,y)$.

Qq- and Quasi-Metric Spaces.

For a relation $s \in \mathcal{Q}\mathbf{Rel}(X,X)$, reflexivity $s(x,x) = 1$ and transitivity $s(x,z) \otimes s(z,y) \sqsubseteq s(x,y)$ can be written more concisely as $s \sqsupseteq \mathbf{1}_X$ and $s \otimes s \sqsubseteq s$. A relation s satisfying both such properties is called a quasi-(pseudo)metric (or a hemi-metric) over X (with values in $\mathcal{Q}$). The "pseudo" prefix stands for the fact that the usual separation property $s(x,y) = 1 \Rightarrow x = y$ needs not hold. As we do not investigate separation here, all metric notions discussed in the rest of the paper are to be understood as implicitly "pseudo".

The following construction generalizes the observational preorders to $\mathcal{Q}$-relations:

Proposition 6.

For all $s \in \mathcal{Q}\mathbf{Rel}(X,X)$, the relations $s^l := s \circ\!\!- s, s^r := s \multimap s \in \mathcal{Q}\mathbf{Rel}(X,X)$ are quasi-metrics and, for $c \in \{l, r\}$, the following hold:

(i.)

$s^c \sqsupseteq s$ iff s is transitive;

(ii.)

$s^c \sqsubseteq s$ iff s is reflexive;

(iii.)

$s^c = s$ iff s is a quasi-metric;

(iv.)

$s^l \otimes s \sqsubseteq s$ and $s \otimes s^r \sqsubseteq s$, that is, the following hold:

$(\forall x,y,z \in X)\ s^l(x,z) \otimes s(z,y) \sqsubseteq s(x,y)$, (left transitivity) $(\forall x,y,z \in X)\ s(x,z) \otimes s^r(z,y) \sqsubseteq s(x,y)$. (right transitivity)

We call the quasi-metrics s^l, s^r the left and right observational quasi-metric of s.

Qq-metrics correspond to $\mathcal{Q}$-relations $s \in \mathcal{Q}\mathbf{Rel}(X,X)$ satisfying transitivity $s \otimes s \sqsubseteq s$ and quasi-reflexivity $s \sqsubseteq \Delta_1 s$ (i.e. $s(x,y) \sqsubseteq s(x,x)$). From transitivity, we deduce that, for a qq-metric s, both $s^l, s^r \sqsupseteq s$ hold, that is, the observational quasi-metrics yield tighter distances than s. This implies that left and right transitivity read as stronger forms of the triangular inequality. In particular, the following alternative characterization of qq-metrics holds:

Proposition 7.

For any quasi-reflexive $s \in \mathcal{Q}\mathbf{Rel}(X,X)$, s is a qq-metric iff there exists a quasi-metric $q \sqsupseteq s$ such that either $s \otimes q \sqsubseteq s$ or $q \otimes s \sqsubseteq s$ holds. Furthermore, when s is a qq-metric, both s^l and s^r are quasi-metrics.

Qq- and Partial Metric Spaces.

Let us now discuss the connection with partial metric spaces. We here consider the non-symmetric variant of the partial metric spaces from [4], called partial quasi-metric spaces (PQM) [24]. As we anticipated, these are metrics p for which the usual reflexivity condition $p(x,x) = 1$ is replaced by the weaker quasi-reflexivity condition

$p(x,x) \sqsupseteq p(x,y)$. However, unlike the qq-metrics just discussed, PQMs satisfy a stronger transitivity condition. When $\mathcal{Q} = [0,+\infty]$ is the Lawvere quantale, this condition reads as

$$p(x,z) + p(z,y) - p(z,z) \sqsupseteq p(x,y).$$

(strong transitivity in $[0,+\infty]$) The idea is that the self-distance of the central term z is "subtracted". For a general quantale $\mathcal{Q}$, this becomes:

$$p(x,z) \otimes (p(z,z) \multimap p(z,y)) \sqsubseteq p(x,y).$$

(strong transitivity)

Define the relations $\Theta^l_s, \Theta^r_s \in \mathcal{Q}\mathbf{Rel}(X,X)$ by $\Theta^l_s(x,y) = s(y,y) \multimap s(x,y)$ and $\Theta^r_s(x,y) = s(x,x) \multimap s(x,y)$. A PQM can be thus more concisely be defined as a relation $s \in \mathcal{Q}\mathbf{Rel}(X,X)$ satisfying $s \sqsubseteq \Delta_1 s$ and $s \otimes \Theta^r_s \sqsubseteq s$. Notice that strong transitivity $s \otimes \Theta^r_s \sqsubseteq s$ looks similar to the right transitivity $s \otimes q^r_s \sqsubseteq s$. Indeed, the following result relates the relations Θ^c_s and s^c:

Proposition 8.

For all $s \in \mathcal{Q}\mathbf{Rel}(X,X)$ and $c \in \{l,r\}$, $s^c \sqsubseteq \Theta^c_s$. Moreover, if s is quasi-reflexive, $\Theta^c_s \sqsubseteq s^c$ holds iff Θ^c_s is a quasi-metric iff s is a partial quasi-metric.

The result above suggests that the partial quasi-metrics can be seen as limit cases of the qq-metrics, namely those for which the quasi-metric $s^r(x,y)$ can be written under the simpler form $\Theta^r_s(x,y) = s(x,x) \multimap s(x,y)$.

Unfortunately, while the standard definition of differential logical relations preserves qq-metrics, it does not preserve partial quasi-metrics: [17, 29] show that the function space constructions lifts PQMs into PQMs only when the monoidal product of the underlying quantales is idempotent (one talks in this case of a partial ultra-metric, since strong transitivity becomes $p(x,z) \wedge p(z,y) \sqsubseteq p(x,y)$). Nevertheless, we will show in Section 7 how one can capture PQMs via a suitable family of logical relations.

4 Differential Logical Relations as Qq-Metrics

In this section we provide a semantic presentation of differential logical relations by defining a cartesian closed category of qq-metrics, this way highlighting the close correspondence between these two notions.

From $\mathcal{Q}$-Relations to Ternary Relations.

While in the previous section we discussed $\mathcal{Q}$-relations, that is, binary relations valued in a quantale $\mathcal{Q}$, the theory of differential logical relations is expressed in terms of ternary relations $\rho \subseteq X \times \mathcal{Q} \times X$. In fact, any such relation $\rho \subseteq X \times \mathcal{Q} \times X$ induces a $\mathcal{Q}$-relation $\widehat{\rho} \in \mathcal{Q}\mathbf{Rel}(X,X)$ defined by

$$\widehat{\rho}(x,y) = \bigvee \{a \in \mathcal{Q} \mid (x,a,y) \in \rho\}.$$

Intuitively, $\widehat{\rho}(x,y)$ is the smallest (recall the inversion of the order) distance between x and y. This correspondence can be made more precise as follows: a ternary relation $\rho \subseteq X \times \mathcal{Q} \times X$ be said to be $\mathcal{Q}$-closed when the following hold:

- $(x,a,y) \in \rho$ and $a' \sqsubseteq a$ implies $(x,a',y) \in \rho$;
- if $(x,a_i,y) \in \rho$, for all $i \in I$, then $(x, \bigvee_{i \in I} a_i, y) \in \rho$.

We will use $\mathcal{Q}$-closedness to derive results in Section 6.

In Appendix A (Lemma 21), we show that the map $\rho \mapsto \widehat{\rho}$ defines a bijection between the $\mathcal{Q}$-closed relations $\rho \subseteq X \times \mathcal{Q} \times X$ and $\mathcal{Q}\mathbf{Rel}(X,X)$. In the sequel, we will identify metrics with their corresponding $\mathcal{Q}$-closed relations.

A Cartesian Closed Category of Qq-Metrics.

We now define a category of qq-metrics. Let us recall notations from Section 2. For sets A and B, we denote the set of functions from A to B by $A \Rightarrow B$; for quantales $\mathcal{Q}$ and $\mathcal{R}$, we denote the set of monotone functions from $\mathcal{Q}$ to $\mathcal{R}$ by $\mathcal{Q} \twoheadrightarrow \mathcal{R}$. Below, we write $f \cdot x$ for the application of $f\colon A \to B$ to $x \in A$, and we suppose that $(-)\cdot(-)$ is left-associative, i.e., $f \cdot x \cdot y$ is an abbreviation of $(f \cdot x) \cdot y$.

The category **Qqm** of qq-metrics is defined as follows:

- objects are triples $X = (\mathcal{Q}_X, |X|, \rho_X)$ consisting of a quantale $\mathcal{Q}_X$, a set $|X|$ and a qq-metric $\rho_X \subseteq |X| \times \mathcal{Q}_X \times |X|$;
- morphisms from X to Y are triples (f, a, f') consisting of functions $f, f'\colon |X| \to |Y|$ and $a\colon |X| \to (\mathcal{Q}_X \twoheadrightarrow \mathcal{Q}_Y)$ such that for all $(x, b, x') \in \rho_X$, we have $(f \cdot x, a \cdot x \cdot b, f' \cdot x') \in \rho_Y$ and $(f \cdot x, a \cdot x \cdot b, f \cdot x') \in \rho_Y$.

The identity morphism on an object X is $(\mathrm{id}_X, i_X, \mathrm{id}_X)$ consisting of the identity function id_X on $|X|$ and a function $i_X\colon |X| \to (\mathcal{Q}_X \twoheadrightarrow \mathcal{Q}_X)$ given by $i_X \cdot x \cdot a = a$. The composition of $(f, a, f')\colon X \to Y$ and $(g, b, g')\colon Y \to Z$ is $(g \circ f, c, g' \circ f')$ where $c\colon |X| \to (\mathcal{Q}_X \twoheadrightarrow \mathcal{Q}_Z)$ is given by $c \cdot x = (b \cdot (f \cdot x)) \circ (a \cdot x)$.

Proposition 9.

The category **Qqm** is cartesian closed.

The cartesian closed structure closely matches the construction of differential logical relations in Section 2. The terminal object $\top$ is $(\{*\}, \{*\}, \rho_\top)$ where $\rho_\top = \{(*, *, *)\}$, and the product of X and Y is $X \times Y = (\mathcal{Q}_X \times \mathcal{Q}_Y, |X| \times |Y|, \rho_{X\times Y})$, where $\rho_{X\times Y}$ is given by

$$((x, y), (a, b), (x', y')) \in \rho_{X\times Y} \iff (x, a, x') \in \rho_X \text{ and } (y, b, y') \in \rho_Y.$$

The exponential $X \Rightarrow Y$ is given by $(|X| \Rightarrow (\mathcal{Q}_X \twoheadrightarrow \mathcal{Q}_Y), |X| \Rightarrow |Y|, \rho_{X\Rightarrow Y})$ where

$$(f, a, f') \in \rho_{X\Rightarrow Y} \iff \text{for all } (x, b, x') \in \rho_X \text{ and } g \in \{f, f'\},\ (f{\cdot}x, a{\cdot}x{\cdot}b, g{\cdot}x') \in \rho_Y.$$

Here, the quantale structure of $\mathcal{Q}_{X\times Y}$ and $\mathcal{Q}_{X\Rightarrow Y}$ are given by the pointwise manner. We give details of the cartesian closed structure in Appendix B.

Example 10.

We define an object $R \in \mathbf{Qqm}$ to be $(\mathbb{R}, [0, +\infty], \rho_R)$, where

$$(x, a, x') \in \rho_R \iff |x - x'| \sqsupseteq a.$$

Observe that the distance $\widehat{\rho}_R : \mathbb{R} \times \mathbb{R} \to [0, +\infty]$ is just the Euclidean distance $\widehat{\rho}_R(x, y) = |y - x|$. For functions $f, g\colon \mathbb{R} \to \mathbb{R}$, an element $a \in \mathcal{Q}_{R\Rightarrow R}$ satisfies $(f, a, g) \in \rho_{R\Rightarrow R}$ if and only if we have $a \cdot x \cdot b \sqsubseteq \bigwedge_{|x-y|\sqsupseteq b} |f \cdot x - g \cdot y|$, i.e., a bounds gaps between outputs of f and g. In particular, we have $(f, \top, f) \in \rho_{R\Rightarrow R}$ if and only if f is a constant function. We note that the largest element $\top \in \mathcal{Q}_{R\Rightarrow R}$ is given by $\top \cdot x \cdot b = 0$.

5 The Fundamental Lemma

In this section we establish the fundamental lemma of differential logical relations for a simply typed lambda calculus Λ_{Real}, by relying on the cartesian closed category **Qqm** of qq-metrics. We then apply this result to measure differences between functions.

Syntax and Set-theoretic Semantics.

Our language Λ_{Real} comprises a type of real numbers and first order functions on $\mathbb{R}$. Let Var be a countably infinite set of variables. We define types and terms as follows:

(type) $\mathsf{A}, \mathsf{B} \coloneqq \mathsf{Real} \mid \mathsf{A} \times \mathsf{B} \mid \mathsf{A} \Rightarrow \mathsf{B}$, (term) $\mathsf{t}, \mathsf{s} \coloneqq \mathsf{x} \in \mathsf{Var} \mid \underline{r} \mid \phi(\mathsf{t}_1, \dots, \mathsf{t}_n) \mid \mathsf{t}\,\mathsf{s} \mid \lambda \mathsf{x} : \mathsf{A}.\, \mathsf{t} \mid \langle \mathsf{t}, \mathsf{s} \rangle \mid \mathsf{fst}(\mathsf{t}) \mid \mathsf{snd}(\mathsf{s})$. Here, r varies over $\mathbb{R}$, and ϕ varies over the set of multi-arity functions on $\mathbb{R}$, namely, ϕ is a function from $\mathbb{R}^n$ to $\mathbb{R}$ for some $n \in \mathbb{N}$. We call n the arity of ϕ, and we denote the arity of ϕ by $\mathrm{ar}(\phi)$. We adopt the standard typing rules given in Figure 1. Below, we denote the set of types by **Type** and the set of closed terms of type A by $\mathbf{T}_{\mathsf{A}}$.

$$\frac{\mathsf{x} : \mathsf{A} \in \Gamma}{\Gamma \vdash \mathsf{x} : \mathsf{A}} \qquad \frac{r \in \mathbb{R}}{\Gamma \vdash \underline{r} : \mathsf{Real}} \qquad \frac{\Gamma \vdash \mathsf{t}_1 : \mathsf{Real} \quad \dots \quad \Gamma \vdash \mathsf{t}_{\mathrm{ar}(\phi)} : \mathsf{Real}}{\Gamma \vdash \phi(\mathsf{t}_1, \dots, \mathsf{t}_{\mathrm{ar}(\phi)}) : \mathsf{Real}} \qquad \frac{\Gamma \vdash \mathsf{t} : \mathsf{A} \Rightarrow \mathsf{B} \quad \Gamma \vdash \mathsf{s} : \mathsf{A}}{\Gamma \vdash \mathsf{t}\,\mathsf{s} : \mathsf{B}}$$

$$\frac{\Gamma, \mathsf{x} : \mathsf{A} \vdash \mathsf{t} : \mathsf{B}}{\Gamma \vdash \lambda \mathsf{x} : \mathsf{A}.\, \mathsf{t} : \mathsf{A} \Rightarrow \mathsf{B}} \qquad \frac{\Gamma \vdash \mathsf{t} : \mathsf{A} \quad \Gamma \vdash \mathsf{s} : \mathsf{B}}{\Gamma \vdash \langle \mathsf{t}, \mathsf{s} \rangle : \mathsf{A} \times \mathsf{B}} \qquad \frac{\Gamma \vdash \mathsf{t} : \mathsf{A} \times \mathsf{B}}{\Gamma \vdash \mathsf{fst}(\mathsf{t}) : \mathsf{A}} \qquad \frac{\Gamma \vdash \mathsf{t} : \mathsf{A} \times \mathsf{B}}{\Gamma \vdash \mathsf{snd}(\mathsf{t}) : \mathsf{B}}$$

Figure 1: Typing Rules.

We denote the standard set theoretic interpretation of Λ_{Real} by $(\!|-|\!)$. (See [25] for example.) To be concrete, the interpretation $(\!|\mathsf{A}|\!)$ of a type A is a set inductively defined by

$$(\!|\mathsf{Real}|\!) = \mathbb{R}, \qquad (\!|\mathsf{A} \times \mathsf{B}|\!) = (\!|\mathsf{A}|\!) \times (\!|\mathsf{B}|\!), \qquad (\!|\mathsf{A} \Rightarrow \mathsf{B}|\!) = (\!|\mathsf{A}|\!) \Rightarrow (\!|\mathsf{B}|\!);$$

and we interpret a term $\mathsf{x}_1 : \mathsf{A}_1, \dots, \mathsf{x}_n : \mathsf{A}_n \vdash \mathsf{t} : \mathsf{B}$ as a function $(\!|\mathsf{t}|\!)$ from $(\!|\mathsf{A}_1|\!) \times \cdots \times (\!|\mathsf{A}_n|\!)$ to $(\!|\mathsf{B}|\!)$. We give the definition of $(\!|\mathsf{t}|\!)$ in Appendix C.

The Fundamental Lemma.

We inductively define a qq-metric space $[\![\mathsf{A}]\!]$ by

$$[\![\mathsf{Real}]\!] = R, \qquad [\![\mathsf{A} \times \mathsf{B}]\!] = [\![\mathsf{A}]\!] \times [\![\mathsf{B}]\!], \qquad [\![\mathsf{A} \Rightarrow \mathsf{B}]\!] = [\![\mathsf{A}]\!] \Rightarrow [\![\mathsf{B}]\!].$$

Here, R is the qq-metric given in Example 10. Below, we simply denote the structure of an object $[\![\mathsf{A}]\!]$ by $(\mathcal{Q}_{\mathsf{A}}, |\mathsf{A}|, \rho_{\mathsf{A}})$. It is straightforward to check that for every type A, we have $|\mathsf{A}| = (\!|\mathsf{A}|\!)$. The qq-metrics $[\![\mathsf{A}]\!]$ are the categorical interpretation of types A, and the following fundamental lemma is derived from the categorical interpretation of Λ_{Real}-terms in **Qqm**.

Theorem 11 (Fundamental Lemma).

Let $\Gamma = (\mathsf{x}_1 : \mathsf{A}_1, \dots, \mathsf{x}_n : \mathsf{A}_n)$ be a typing context. For every term $\Gamma \vdash \mathsf{t} : \mathsf{A}$, and for every $(x, a, x') \in \rho_{\mathsf{A}_1 \times \cdots \times \mathsf{A}_n}$, we have

$$((\!|\mathsf{t}|\!) \cdot x, \{\!|\mathsf{t}|\!\} \cdot x \cdot a, (\!|\mathsf{t}|\!) \cdot x') \in \rho_{\mathsf{A}}$$

where we inductively define $\{\!|\mathsf{t}|\!\} \in \mathcal{Q}_{\mathsf{A}_1 \times \cdots \times \mathsf{A}_n \Rightarrow \mathsf{B}}$ as follows:

■

We define $\{\!|\mathsf{x}_i|\!\} \cdot (x_1, \dots, x_n) \cdot (a_1, \dots, a_n)$ to be a_i.

■

We define $\{\!|\underline{r}|\!\} \cdot x \cdot a$ to be $\underline{0}$.

■

We define $\{\!|\phi(\mathsf{t}_1, \dots, \mathsf{t}_n)|\!\} \cdot x \cdot a$ to be $\phi^{\mathrm{d}}((\!|\mathsf{t}_1|\!) \cdot x, \dots, (\!|\mathsf{t}_n|\!) \cdot x, \{\!|\mathsf{t}_1|\!\} \cdot x \cdot a, \dots, \{\!|\mathsf{t}_n|\!\} \cdot x \cdot a)$ where we define $\phi^{\mathrm{d}} \colon \mathbb{R}^n \times [0, +\infty]^n \to [0, +\infty]$ by

$$\phi^{\mathrm{d}}(y_1, \dots, y_n, b_1, \dots, b_n) = \bigwedge_{|y_1 - z_1| \sqsupseteq b_1} \cdots \bigwedge_{|y_n - z_n| \sqsupseteq b_n} |\phi(y_1, \dots, y_n) - \phi(z_1, \dots, z_n)|.$$

■

We define $\{\!|\mathsf{t}\,\mathsf{s}|\!\} \cdot x \cdot a$ to be $\{\!|\mathsf{t}|\!\} \cdot x \cdot a \cdot ((\!|\mathsf{s}|\!) \cdot x) \cdot (\{\!|\mathsf{s}|\!\} \cdot x \cdot a)$.

■

We define $\{\!|\lambda \mathsf{x} : \mathsf{A}.\, \mathsf{t}|\!\} \cdot (x_1, \ldots, x_n) \cdot (a_1, \ldots, a_n) \cdot y \cdot b$ to be $\{\!|\mathsf{t}|\!\} \cdot (x_1, \ldots, x_n, y) \cdot (a_1, \ldots, a_n, b)$.

■

We define $\{\!|\langle \mathsf{t}, \mathsf{s}\rangle|\!\} \cdot x \cdot a$ to be $(\{\!|\mathsf{t}|\!\} \cdot x \cdot a, \{\!|\mathsf{s}|\!\} \cdot x \cdot a)$.

■

We define $\{\!|\mathsf{fst}(\mathsf{t})|\!\} \cdot x \cdot a$ to be the first component of $\{\!|\mathsf{t}|\!\} \cdot x \cdot a$.

■

We define $\{\!|\mathsf{snd}(\mathsf{t})|\!\} \cdot x \cdot a$ to be the second component of $\{\!|\mathsf{t}|\!\} \cdot x \cdot a$.

The fundamental lemma is a way to compositionally reason about distances.

Example 12.

Let us fix a positive real number ϵ. We define $D_\epsilon \colon (\mathbb{R} \Rightarrow \mathbb{R}) \to (\mathbb{R} \Rightarrow \mathbb{R})$ by

$$D_\epsilon = (\!|\lambda \mathsf{f} : \mathsf{Real} \Rightarrow \mathsf{Real}.\, \lambda \mathsf{x} : \mathsf{Real}.\, \mathrm{df}_\epsilon(f(\mathrm{ad}_\epsilon(\mathsf{x})), f(\mathsf{x}))|\!)$$

where $\mathrm{df}_\epsilon(x, y) = \frac{x-y}{\epsilon}$ and $\mathrm{ad}_\epsilon(x) = x + \epsilon$. For $f \colon \mathbb{R} \to \mathbb{R}$ and $x \in \mathbb{R}$, $D_\epsilon \cdot f \cdot x$ calculates an approximation of the derivative of f at x:

$$D_\epsilon \cdot f \cdot x = \frac{f(x+\epsilon) - f(x)}{\epsilon}.$$

By the fundamental lemma, we obtain $(D_\epsilon, E_\epsilon, D_\epsilon) \in \rho_{(R \Rightarrow R) \Rightarrow (R \Rightarrow R)}$ where E_ϵ is a function from $|R \Rightarrow R|$ to $\mathcal{Q}_{R \Rightarrow R} \nrightarrow \mathcal{Q}_{R \Rightarrow R}$ given by

$$E_\epsilon \cdot f \cdot a = \lambda x : \mathbb{R}.\, \lambda b : [0, +\infty].\, \frac{a \cdot (x+\epsilon) \cdot b + a \cdot x \cdot b}{\epsilon}.$$

We note that E_ϵ depends on our choice of term denoting D_ϵ. In Example 15, we will observe that $(\mathrm{id}_\mathbb{R}, a, \sin)$ is an element of $\rho_{R \Rightarrow R}$ where $\mathrm{id}_\mathbb{R}$ is the identity function on $\mathbb{R}$, and $a \in \mathcal{Q}_{R \Rightarrow R}$ is given by $a \cdot x \cdot b = |x - \sin(x)| + b$. By applying $(D_\epsilon, E_\epsilon, D_\epsilon)$ to $(\mathrm{id}_\mathbb{R}, a, \sin)$, we obtain $(D_\epsilon \cdot \mathrm{id}_\mathbb{R}, a', D_\epsilon \cdot \sin) \in \rho_R$ where

$$a' \cdot x \cdot b = \frac{|x + \epsilon - \sin(x+\epsilon)| + |x - \sin(x)| + 2b}{\epsilon}.$$

From this, we see that the distance between $D_\epsilon \cdot \mathrm{id}_\mathbb{R} \cdot 0$ and $D_\epsilon \cdot \sin \cdot 0$ is bounded by $\frac{|\epsilon - \sin(\epsilon)|}{\epsilon}$. We note that a' is not the exact distance between $D_\epsilon \cdot \mathrm{id}_\mathbb{R}$ and $D_\epsilon \cdot \sin$. For example, while $|D_{0.1} \cdot \mathrm{id}_\mathbb{R} \cdot 0 - D_{0.1} \cdot \sin \cdot 0.1| \approx 0.01$, we have $a' \cdot 0 \cdot 0.1 \approx 2$. This gap stems in the fact that $(D_\epsilon, E_\epsilon, D_\epsilon)$ takes all functions into account and cannot exploit continuity of specific functions.

6 Quasi-Metric Logical Relations

As described in Section 3, any qq-metric gives rise to left and right observational quasi-metrics. In this section, we introduce a class of logical relations γ_A that capture the left observational quasi-metric associated to ρ_A. We will then show how such relations can be used to derive over-approximations of distances between functions.

For a type A, we define $\gamma_\mathsf{A} \subseteq |\mathsf{A}| \times \mathcal{Q}_\mathsf{A} \times |\mathsf{A}|$ by induction on A as follows:

$(x, a, x') \in \gamma_{\mathsf{Real}} \iff |x - x'| \sqsupseteq a$, $(f, a, f') \in \gamma_{\mathsf{A} \Rightarrow \mathsf{B}} \iff$ for all $(x, b, x') \in \rho_\mathsf{A}$, $(f \cdot x, a \cdot x \cdot b, f' \cdot x) \in \gamma_\mathsf{B}$, and for all $(f', b, f') \in \rho_{\mathsf{A} \Rightarrow \mathsf{B}}$, $(f, a \otimes b, f) \in \rho_{\mathsf{A} \Rightarrow \mathsf{B}}$, $((x, y), (a, b), (x', y')) \in \gamma_{\mathsf{A} \times \mathsf{B}} \iff (x, a, x') \in \gamma_\mathsf{A}$ and $(y, b, y') \in \gamma_\mathsf{B}$. We give some explanation on the definition of $\gamma_{\mathsf{A} \Rightarrow \mathsf{B}}$. The definition consists of two conditions. The first condition means that, if (f, a, f') is an element of $\gamma_{\mathsf{A} \Rightarrow \mathsf{B}}$, then the distance $a \cdot x \cdot b$ over-approximates the distance between f and f' at the same point x (rather than on distinct points, as is the case for the relation $\rho_{\mathsf{A} \Rightarrow \mathsf{B}}$). The second condition means that a also over-approximates the gap between the self-distance of f' and the self-distance of f.

Recall that ρ^l_{A} is the quasi-metric representing the left observational quasi-metrics associated with the qq-metric ρ_{A}. In Appendix D, we give some auxiliary lemmas and proofs for Proposition 13 and Theorem 14.

We can use Proposition 13 to over-approximate ρ-distances in terms of γ-distances and the left observational quasi-metric. Let us sketch our idea. First, thanks to Proposition 13 and Proposition 6, we can exploit left-transitivity to pass from a γ-distance between t and s and a self-ρ-distance of s to a ρ-distance between t and s:

$(\!|\mathsf{t}|\!), a, (\!|\mathsf{s}|\!)) \in \gamma_{\mathsf{A}}$ and $((\!|\mathsf{s}|\!), b, (\!|\mathsf{s}|\!)) \in \rho_{\mathsf{A}} \implies ((\!|\mathsf{t}|\!), a \otimes b, (\!|\mathsf{s}|\!)) \in \rho_{\mathsf{A}}$. (left transitivity)
Second, thanks to the fundamental lemma, we can always obtain a ρ-distance by summing a γ-distance with the self-distance $\{\!|\mathsf{s}|\!\}$:

$((\!|\mathsf{t}|\!), a, (\!|\mathsf{s}|\!)) \in \gamma_{\mathsf{A}} \implies ((\!|\mathsf{t}|\!), a \otimes \{\!|\mathsf{s}|\!\}, (\!|\mathsf{s}|\!)) \in \rho_{\mathsf{A}}$. ($\rho \sqsupseteq \gamma \otimes$ self-ρ) The following result exploits this last idea to bound the distance between two functions f and g by summing the "vertical distance" between f and g (that is, the distance of $f(x)$ and $g(x)$ for some fixed x) with an approximation of the self-distances of f and g:

Example 15.

Let $\mathrm{id}_{\mathbb{R}}$ be the identity function on $\mathbb{R}$. By the fundamental lemma with a simple calculation, we obtain $(\mathrm{id}_{\mathbb{R}}, a', \mathrm{id}_{\mathbb{R}}) \in \rho_{R \Rightarrow R}$ and $(\sin, a', \sin) \in \rho_{R \Rightarrow R}$ where $a' \cdot x \cdot b = b$. By Theorem 14, $a \in \mathcal{Q}_{R \Rightarrow R}$ given by $a \cdot x \cdot b = |x - \sin(x)|$ satisfies $(\mathrm{id}_{\mathbb{R}}, a \otimes a', \sin) \in \rho_{R \Rightarrow R}$. To be concrete, $(a \otimes a') \cdot x \cdot b = |x - \sin(x)| + b$, which means that the distance between x and $\sin(y)$ is small when x and y are close to 0.

▶ **Remark 16.**

Due to asymmetry in the definition of the exponential $X \Rightarrow Y$ in **Qqm**, it is not clear how to capture the right observational quasi-metrics in a similar manner. However, we will see that right observational quasi-metrics can be captured by partial metric logical relations introduced in the next section.

7 Partial Metric Logical Relations

As discussed in Section 3, the qq-metrics ρ_{A} are not, in general, partial metrics. In this section we introduce a family of differential logical relations $(\eta_{\mathsf{A}})_{\mathsf{A} \in \mathsf{Types}}$ that defines a class of partial quasi-metrics over Λ_{Real}. The fundamental (indeed, the only) difference with respect to the family ρ_{A} is, as it may be expected, in the case of the function type.

For any type A, we define $\eta_{\mathsf{A}} \subseteq |\mathsf{A}| \times \mathcal{Q}_{\mathsf{A}} \times |\mathsf{A}|$ by induction on A as follows:

$(x, a, x') \in \eta_{\mathsf{Real}} \iff |x - x'| \sqsupseteq a$, $(f, a, f') \in \eta_{\mathsf{A} \Rightarrow \mathsf{B}} \iff$ there are $a_1, a_2 \in \mathcal{Q}_{\mathsf{A} \Rightarrow \mathsf{B}}$ such that $a_1 \otimes a_2 \sqsupseteq a$ and for all $(x, b, x') \in \eta_{\mathsf{A}}$, $(f \cdot x, a_1 \cdot x \cdot b, f \cdot x') \in \eta_{\mathsf{B}}$ and $(f \cdot x', a_2 \cdot x \cdot b, f' \cdot x') \in \eta_{\mathsf{B}}$, $((x, y), (a, b), (x', y')) \in \eta_{\mathsf{A} \times \mathsf{B}} \iff (x, a, x') \in \eta_{\mathsf{A}}$ and $(y, b, y') \in \eta_{\mathsf{B}}$. The idea of the definition of $\eta_{\mathsf{A} \Rightarrow \mathsf{B}}$ is that if $(f, a, f') \in \eta_{\mathsf{A} \Rightarrow \mathsf{B}}$, then a must be larger than or equal to the sum of the self-distance of f and of the "vertical" distances between f and f'. The following result shows that the relations η_{A} define partial quasi-metrics on all types.

For $(f, a, f') \in \eta_{\mathsf{A} \Rightarrow \mathsf{B}}$, we call a pair $a_1, a_2 \in \mathcal{Q}_{\mathsf{A} \Rightarrow \mathsf{B}}$ satisfying the condition in the definition of $(f, a, f') \in \eta_{\mathsf{A} \Rightarrow \mathsf{B}}$ a decomposition of $(f, a, f') \in \eta_{\mathsf{A} \Rightarrow \mathsf{B}}$.

Proposition 17.

For all types A:

■

For any type A, the relation ρ_{A} is $\mathcal{Q}$-closed. In particular, for all $(x, a, x') \in \eta_{\mathsf{A}}$, the set of decompositions of $(x, a, x') \in \eta_{\mathsf{A}}$ is a complete lattice.

■

If $(x, a, x') \in \eta_{\mathsf{A}}$, then $(x, a, x) \in \eta_{\mathsf{A}}$.

■

If $(x,a,z) \in \eta_{\mathsf{A}}$ and $(z,b,y) \in \eta_{\mathsf{A}}$, then there exists $c_1, c_2 \in \mathcal{Q}_{\mathsf{A}}$ such that $a \otimes b \sqsubseteq c_1 \otimes c_2$, $(z, c_1, z) \in \eta_{\mathsf{A}}$ and $(x, c_2, y) \in \eta_{\mathsf{A}}$. In particular, $(x, a \otimes (c_1 \multimap b), y) \in \eta_{\mathsf{A}}$.

By adapting the definition of γ_{A} from Section 6, we can capture the left observational quasi-metrics ρ^l_{A} associated with the partial quasi-metrics η_{A}. Moreover, by Proposition 8, the right observational quasi-metric ρ^r_{A} satisfies $(x, \widehat{\eta_{\mathsf{A}}}(x,x) \multimap a, y) \in \rho^r_{\mathsf{A}} \iff (x,a,y) \in \eta_{\mathsf{A}}$. Thanks to this, we can capture this quasi-metrics via the logical relations $\delta_{\mathsf{A}} \subseteq |\mathsf{A}| \times \mathcal{Q}_{\mathsf{A}} \times |\mathsf{A}|$ defined by induction on A, letting the base and product case being defined as for γ_{A}, and the function case being as follows:

$(f,a,f') \in \delta_{\mathsf{A}\Rightarrow\mathsf{B}} \iff$ for all $(f,b,f) \in \eta_{\mathsf{A}\Rightarrow\mathsf{B}}$, $(f, a \otimes b, f') \in \eta_{\mathsf{A}\Rightarrow\mathsf{B}}$.

Proposition 13.
For every type A, we have $\rho^l_{\mathsf{A}} = \gamma_{\mathsf{A}}$.

Theorem 14.
Let A be a type. For any $f, f' \in |\mathsf{A} \Rightarrow \mathsf{Real}|$ and any $a, a' \in \mathcal{Q}_{\mathsf{A}\Rightarrow\mathsf{Real}}$, if

- $|f \cdot x - f' \cdot x| \sqsupseteq a \cdot x \cdot b$ for all $(x, b, x') \in \rho_{\mathsf{A}}$; and
- $(f, a', f) \in \rho_{\mathsf{A}\Rightarrow\mathsf{Real}}$ and $(f', a', f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{Real}}$,

then $(f, a \otimes a', f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{Real}}$.

Proposition 18.
For every type A, we have $\rho^r_{\mathsf{A}} = \delta_{\mathsf{A}}$.

Appendix A Proofs for Section 3

Proof of Proposition 7.
Suppose there exists a quasi-metric $q \sqsupseteq s$ such that $s \otimes q \sqsubseteq s$ holds. Then $s \otimes s \sqsubseteq s \otimes q \sqsubseteq s$, so s is transitive. A similar argument works if q is such that $q \otimes s \sqsubseteq s$. Conversely, if s is a qq-metric it is enough to let $q := s^r$ and use Proposition 6 (iv). ◀

Proof of Proposition 8.
We only argue for $c = r$, the other case being similar. From $s^r(x,y) = \bigwedge_z s(z,x) \multimap s(z,y) \sqsubseteq s(x,x) \multimap s(x,y) = (\Theta^r_s)(x,y)$ we deduce that $s^r \sqsubseteq \Theta^r_s$. The converse direction $s^r \sqsupseteq \Theta^r_s$ corresponds to showing that $s(z,x) \otimes (s(x,x) \multimap s(x,y)) \sqsubseteq s(z,y)$, which holds iff s is a partial quasi-metric. We have thus shown that s is a PQM iff $\Theta^r_s = s^r$. This also implies that, if s is a PWM, Θ^r_s is a quasi-metric. Finally, suppose Θ^r_s is a quasi-metric. By quasi-reflexivity, and the divisibility of $\mathcal{Q}$, we have that $s(x,z) = s(x,x) \otimes (s(x,x) \multimap s(x,z))$. We then have $s(x,z) \otimes (s(z,z) \multimap s(z,y)) = s(x,x) \otimes (s(x,x) \multimap s(x,z)) \otimes (s(z,z) \multimap s(z,y)) \sqsubseteq s(x,y)$, so s is a partial quasi-metric. ◀

Lemma 21.
The map $\rho \mapsto \widehat{\rho}$ defines a bijection between the $\mathcal{Q}$-closed relations $\rho \subseteq X \times \mathcal{Q} \times X$ and $\mathcal{Q}\mathbf{Rel}(X,X)$.

Proof.
Let ρ, τ be closed and let $\widehat{\rho}(x,y) = \widehat{\tau}(x,y)$. Observe that, for all $x, y \in X$, by $\mathcal{Q}$-closure we have $(x, \widehat{\rho}(x,y), y) \in \rho$. Suppose now that $(x,a,y) \in \tau$, then $a \sqsubseteq \widehat{\tau}(x,y) = \widehat{\rho}(x,y)$, and from $(x, \widehat{\rho}(x,y), y) \in \rho$ and $a \sqsubseteq \widehat{\rho}(x,y)$ we deduce $(x,a,y) \in \rho$. By a similar argument we can also prove that $(x,a,y) \in \rho$ implies $(x,a,y) \in \tau$, so in the end $\rho = \tau$. We conclude then that the map $\rho \mapsto \widehat{\rho}$ is injective. For surjectivity, observe that any $s \in \mathcal{Q}\mathbf{Rel}(X,X)$ induces a relation $(x,a,y) \in \rho^s$ iff $a \sqsubseteq s(x,y)$, so that $s = \widehat{\rho^s}$. ◀

Appendix B Details of Cartesian Closed Structure

The first projection from $X \times Y$ to Y is given by $(\mathrm{proj}_{X,Y}, \varpi_{X,Y}, \mathrm{proj}_{X,Y})$ consisting of the first projection $\mathrm{proj}_{X,Y} \colon |X| \times |Y| \to |X|$ and $\varpi_{X,Y} \colon |X| \times |Y| \to (\mathcal{Q}_X \times \mathcal{Q}_Y \multimap \mathcal{Q}_X)$ given by $\varpi_{X,Y} \cdot (x,y) \cdot (a,b) = a$. The second projection is given in the same manner. The tupling of $(f,a,f') \colon Z \to X$ and $(g,b,g') \colon Z \to Y$ is $(\langle f,g\rangle, \langle a,b\rangle, \langle f',g'\rangle)$ where $\langle f,g\rangle \colon |Z| \to |X| \times |Y|$ and $\langle a,b\rangle \colon |Z| \to (\mathcal{Q}_Z \multimap \mathcal{Q}_X \times \mathcal{Q}_Y)$ are the tupling of f,g and a,b respectively, that is, $\langle f,g\rangle \cdot z$ is defined to be $(f \cdot z, g \cdot z)$, and $\langle a,b\rangle \cdot z \cdot c$ is defined to be $(a \cdot z \cdot c, a \cdot z \cdot c)$. The currying of $(f,a,f') \colon Z \times X \to Y$ is $(f^\wedge, a^\wedge, f'^\wedge)$ where $f^\wedge \colon |Z| \to (|X| \Rightarrow |Y|)$ and $f'^\wedge \colon |Z| \to (|X| \Rightarrow |Y|)$ are the currying of $f \colon |Z| \times |X| \to |Y|$ and $f' \colon |Z| \times |X| \to |Y|$, and $a^\wedge \colon |Z| \to (\mathcal{Q}_Z \multimap \mathcal{Q}_{X \Rightarrow Y})$ is the currying of $a \colon |Z| \times |X| \to (\mathcal{Q}_{Z \times X} \multimap \mathcal{Q}_Y)$, namely, $a^\wedge \cdot z \cdot a \cdot x \cdot b$ is defined to be $a \cdot (z,x) \cdot (a,b)$. The evaluation morphism $(\mathrm{eval}_{X,Y}, \varepsilon_{X,Y}, \mathrm{eval}_{X,Y}) \colon (X \Rightarrow Y) \times X \to Y$ consists of the evaluation function $\mathrm{eval}_{X,Y} \colon (|X| \Rightarrow |Y|) \times |X| \to |Y|$ and $\varepsilon_{X,Y}$ is given by $\varepsilon_{X,Y} \cdot (a,b) \cdot (f,x) = a \cdot x \cdot b$.

Appendix C Definitions and Proofs for Section 5

In Figure 4, we define the function $(\!|\mathsf{t}|\!)$ by induction on the type derivation of t.

$(\!|\mathsf{x}_i|\!) \cdot (x_1,\dots,x_n) = x_i$, $(\!|\underline{r}|\!) \cdot (x_1,\dots,x_n) = r$, $(\!|\phi(\mathsf{t}_1,\dots,\mathsf{t}_{\mathrm{ar}(\phi)})|\!) \cdot (x_1,\dots,x_n)$ $= \phi((\!|\mathsf{t}_1|\!)\cdot(x_1,\dots,x_n),\dots,(\!|\mathsf{t}_{\mathrm{ar}(\phi)}|\!)\cdot(x_1,\dots,x_n))$, $(\!|\mathsf{t}\,\mathsf{s}|\!)\cdot(x_1,\dots,x_n) = (\!|\mathsf{t}|\!)\cdot(x_1,\dots,x_n)\cdot$ $((\!|\mathsf{s}|\!)\cdot(x_1,\dots,x_n))$, $(\!|\lambda\mathsf{x}\colon\mathsf{A}.\,\mathsf{t}|\!)\cdot(x_1,\dots,x_n)\cdot y = (\!|\mathsf{t}|\!)\cdot(x_1,\dots,x_n,y)$, $(\!|\mathsf{fst}(\mathsf{t})|\!)\cdot(x_1,\dots,x_n)$ $=$ the first component of $(\!|\mathsf{t}|\!)\cdot(x_1,\dots,x_n)$, $(\!|\mathsf{snd}(\mathsf{t})|\!)\cdot(x_1,\dots,x_n) =$ the second component of $(\!|\mathsf{t}|\!)\cdot$ $(x_1,\dots,x_n)$, $(\!|\langle\mathsf{t},\mathsf{s}\rangle|\!) \cdot (x_1,\dots,x_n) = ((\!|\mathsf{t}|\!) \cdot (x_1,\dots,x_n), (\!|\mathsf{s}|\!) \cdot (x_1,\dots,x_n))$. Figure 4: Set Theoretic Denotation of Terms.

Proof of Theorem 11.

The triple $((\!|\mathsf{t}|\!), \{\!|\mathsf{t}|\!\}, (\!|\mathsf{t}|\!))$ is the interpretation of $\Gamma \vdash \mathsf{t} : \mathsf{A}$ in the cartesian closed category **Qqm** where we interpret $\Gamma \vdash \phi(\mathsf{t}_1,\dots,\mathsf{t}_n)$ by

$$[\![\mathsf{A}_1]\!] \times \cdots \times [\![\mathsf{A}_n]\!] \xrightarrow{\langle [\![\mathsf{t}_1]\!],\dots,[\![\mathsf{t}_n]\!]\rangle} R \times \cdots \times R \xrightarrow{(\phi,\phi^{\mathrm{d}},\phi)} R.$$

The statement follows from that $((\!|\mathsf{t}|\!), \{\!|\mathsf{t}|\!\}, (\!|\mathsf{t}|\!))$ is a morphism from $[\![\mathsf{A}_1]\!] \times \cdots \times [\![\mathsf{A}_n]\!]$ to $[\![\mathsf{A}]\!]$ in **Qqm**. ◀

Appendix D Proofs for Section 6

Let us introduce a notation. For a type A and $x \in |\mathsf{A}|$, we write $\lceil x \rceil \in \mathcal{Q}_\mathsf{A}$ for $\widehat{\rho_\mathsf{A}}(x,x)$, i.e., $\lceil x \rceil$ is equal to $\sup\{a \in \mathcal{Q}_\mathsf{A} \mid (x,a,x) \in \rho_\mathsf{A}\}$. Since ρ_A is closed under supremum, for any $x \in |\mathsf{A}|$, we have $(x, \lceil x \rceil, x) \in \rho_\mathsf{A}$.

Lemma 22.

For every type A and for every $x, x' \in |\mathsf{A}|$, if $(x, \lceil x \rceil, x') \in \rho_\mathsf{A}$, then $x = x'$.

Proof.

By induction on A. It is straightforward to check the case Real and the case $\mathsf{A} \times \mathsf{B}$. For the case $\mathsf{A} \Rightarrow \mathsf{B}$, if $(f, \lceil f \rceil, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$, then for any $x \in |\mathsf{A}|$, we have

$$(f \cdot x, \lceil f \rceil \cdot x \cdot \lceil x \rceil, f' \cdot x) \in \rho_\mathsf{A}.$$

Here, by the induction hypothesis,

$\lceil f \rceil \cdot x \cdot \lceil x \rceil = \sup\{a \in \mathcal{Q}_\mathsf{B} \mid \text{for all } (x, \lceil x \rceil, x') \in \rho_\mathsf{A},\ (f \cdot x, a, f \cdot x') \in \rho_\mathsf{B}\}$ $= \sup\{a \in \mathcal{Q}_\mathsf{B} \mid (f \cdot x, a, f \cdot x) \in \rho_\mathsf{B}\} = \lceil f \cdot x \rceil$. Hence, $f' \cdot x = f \cdot x$. ◀

Lemma 23.

For any type A and B, if $f \in |\mathsf{A} \Rightarrow \mathsf{B}|$ and $x \in |\mathsf{A}|$, then $\lceil f \rceil \cdot x \cdot \lceil x \rceil = \lceil f \cdot x \rceil$.

Proof.

This is shown in the proof of Lemma 22. ◀

Lemma 24.

For every type A, if $(x, a \otimes \lceil x' \rceil, x') \in \rho_\mathsf{A}$, then $(x, a, x') \in \gamma_\mathsf{A}$.

Proof.

By induction on A. The only non-trivial case is $\mathsf{A} \Rightarrow \mathsf{B}$. If $(f, a \otimes \lceil f' \rceil, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$, then for any $(x, b, x') \in \rho_{\mathsf{A}}$, since $(x, b \vee \lceil x \rceil, x) \in \rho_{\mathsf{A}}$, we obtain

$$(f \cdot x, (a \cdot x \cdot (b \vee \lceil x \rceil)) \otimes (\lceil f' \rceil \cdot x \cdot (b \vee \lceil x \rceil)), f' \cdot x) \in \rho_{\mathsf{B}}.$$

It follows from monotonicity of a and Lemma 23 that we have

$$(f \cdot x, (a \cdot x \cdot b) \otimes (\lceil f' \rceil \cdot x \cdot \lceil x \rceil), f' \cdot x) = (f \cdot x, (a \cdot x \cdot b) \otimes \lceil f' \cdot x \rceil, f' \cdot x) \in \rho_{\mathsf{B}}.$$

By the induction hypothesis, we conclude $(f \cdot x, a \cdot x \cdot b, f' \cdot x) \in \gamma_{\mathsf{B}}$. For any $(f', b, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$, since $b \sqsubseteq \lceil f' \rceil$, it follows from $(f, a \otimes \lceil f' \rceil, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$ that $(f, a \otimes b, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$. By left-quasi-reflexivity, we obtain $(f, a \otimes b, f) \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$. ◀

Proof of Proposition 13.

We first show that γ_{A} is a subset of ρ^l_{A} by induction on A. It is straightforward to check the case Real and the case $\mathsf{A} \times \mathsf{B}$. We check the case $\mathsf{A} \Rightarrow \mathsf{B}$. Let (f, a, f') be an element of $\gamma_{\mathsf{A}\Rightarrow\mathsf{B}}$, and let (f', a', f'') be an element of $\rho_{\mathsf{A}\Rightarrow\mathsf{B}}$. We show that $(f, a \otimes a', f'')$ is an element of $\rho_{\mathsf{A}\Rightarrow\mathsf{B}}$. For any $(x, b, x') \in \rho_{\mathsf{A}}$, since $(x, b, x) \in \rho_{\mathsf{A}}$, we have

$$(f \cdot x, a \cdot x \cdot b, f' \cdot x) \in \gamma_{\mathsf{B}}, \qquad (f' \cdot x, a' \cdot x \cdot b, f'' \cdot x') \in \rho_{\mathsf{B}}.$$

Hence, by the induction hypothesis, we see that $(f \cdot x, (a \otimes a') \cdot x \cdot b, f'' \cdot x')$ is an element of ρ_{B}. It remains to check that $(f \cdot x, (a \otimes a') \cdot x \cdot b, f \cdot x')$ is an element of ρ_{B}. Since $(f', a', f'') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$, we have $(f', a', f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$. Then, by the definition of $\gamma_{\mathsf{A}\Rightarrow\mathsf{B}}$, we obtain $(f, a \otimes a', f) \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$. Hence, $(f \cdot x, (a \otimes a') \cdot x \cdot b, f \cdot x')$ is an element of ρ_{B}. We next show that ρ^l_{A} is a subset of γ_{A}. Again, it is straightforward to check the case Real and the case $\mathsf{A} \times \mathsf{B}$. We check the case $\mathsf{A} \Rightarrow \mathsf{B}$. Let (f, a, f') be an element of $\rho^l_{\mathsf{A}\Rightarrow\mathsf{B}}$, and let (x, b, x') be an element of ρ_{A}. Since $(f, a \otimes \lceil f' \rceil, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$ and $(x, \lceil x \rceil, x) \in \rho_{\mathsf{A}}$, we obtain

$$(f \cdot x, (a \cdot x \cdot \lceil x \rceil) \otimes (\lceil f' \rceil \cdot x \cdot \lceil x \rceil), f' \cdot x) = (f \cdot x, (a \cdot x \cdot \lceil x \rceil) \otimes \lceil f' \cdot x \rceil, f' \cdot x) \in \rho_{\mathsf{B}}$$

Hence, by Lemma 24, $(f \cdot x, a \cdot x \cdot \lceil x \rceil, f' \cdot x)$ is an element of ρ_{B}. Since a is monotone, for any $(x, b, x') \in \rho_{\mathsf{A}}$, we have $(f \cdot x, a \cdot x \cdot b, f' \cdot x) \in \rho_{\mathsf{B}}$. If $(f', b, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$, then by the definition of $\rho^l_{\mathsf{A}\Rightarrow\mathsf{B}}$, we obtain $(f, a \otimes b, f) \in \rho_{\mathsf{A}\Rightarrow\mathsf{B}}$. ◀

Proof of Theorem 14.

By the definition of $\gamma_{\mathsf{A}\Rightarrow\mathsf{Real}}$, we obtain

$$(f, a \otimes (\lceil f' \rceil \multimap \lceil f \rceil), f') \in \gamma_{\mathsf{A}\Rightarrow\mathsf{Real}}.$$

Therefore, it follows from Proposition 13 that

$$(f, a \otimes (\lceil f' \rceil \multimap \lceil f \rceil) \otimes \lceil f' \rceil, f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{Real}}.$$

Since $(f, a', f) \in \rho_{\mathsf{A}\Rightarrow\mathsf{Real}}$ and $(f', a', f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{Real}}$, we have $a' \sqsubseteq (\lceil f' \rceil \multimap \lceil f \rceil) \otimes \lceil f' \rceil$. Hence, $(f, a \otimes a', f') \in \rho_{\mathsf{A}\Rightarrow\mathsf{Real}}$. ◀

Appendix E Proofs and Observations for Section 7

Proof of Proposition 17.

We can check the first clause and the second clause by induction on A. We only prove the third, more delicate, statement. By induction on A, we show that there is a map $\varphi_{\mathsf{A}} \colon \mathcal{Q}_{\mathsf{A}} \times \mathcal{Q}_{\mathsf{A}} \Rightarrow \mathcal{Q}_{\mathsf{A}} \times \mathcal{Q}_{\mathsf{A}}$ such that if $(x, a, z) \in \eta_{\mathsf{A}}$ and $(z, b, y) \in \eta_{\mathsf{A}}$, then $(c_1, c_2) = \varphi_{\mathsf{A}}(a, b)$ satisfies the required conditions. For the base case, we define $\varphi_{\mathsf{Real}}(a, b) = (0, a + b)$. For the case $\mathsf{A} = (\mathsf{B} \Rightarrow \mathsf{C})$, let $(a_1, a_2) \in \mathcal{Q}_{\mathsf{B}\Rightarrow\mathsf{C}} \times \mathcal{Q}_{\mathsf{B}\Rightarrow\mathsf{C}}$ and $(b_1, b_2) \in \mathcal{Q}_{\mathsf{B}\Rightarrow\mathsf{C}} \times \mathcal{Q}_{\mathsf{B}\Rightarrow\mathsf{C}}$ be the greatest decompositions of a and b, respectively. We define $\varphi_{\mathsf{B}\Rightarrow\mathsf{C}}(a, b)$ by

$$\varphi_{\mathsf{B}\Rightarrow\mathsf{C}}(a,b) = (a_1 \otimes k, b_1 \otimes l)$$

where $(k \cdot w \cdot d, l \cdot w \cdot d) = \varphi_{\mathsf{C}}(a_2 \cdot w \cdot d, b_2 \cdot w \cdot d)$. Below, we write (c_1, c_2) for $\varphi_{\mathsf{B}\Rightarrow\mathsf{C}}(a,b)$. Let us check that $\varphi_{\mathsf{B}\Rightarrow\mathsf{C}}$ is a witness.

■

We first show that $(z, c_1, z) \in \eta_{\mathsf{B}\Rightarrow\mathsf{C}}$. For any $(w, d, w') \in \eta_{\mathsf{B}}$, we have $(x\cdot w', a_2\cdot w\cdot d, z\cdot w') \in \eta_{\mathsf{C}}$, (1) $(z\cdot w, b_1\cdot w\cdot d, z\cdot w') \in \eta_{\mathsf{C}}$, (2) $(z\cdot w', b_2\cdot w\cdot d, y\cdot w') \in \eta_{\mathsf{C}}$. (3) Then, by applying the induction hypothesis to (1) and (3), we obtain

$$(z \cdot w', k \cdot w \cdot d, z \cdot w') \in \eta_{\mathsf{C}}.$$

(4) By (2) and (4), we obtain $(z, c_1, z) \in \eta_{\mathsf{B}\Rightarrow\mathsf{C}}$.

■

We next show that $(x, c_2, y) \in \eta_{\mathsf{B}\Rightarrow\mathsf{C}}$. For any $(w, d, w') \in \eta_{\mathsf{B}}$, we have $(x\cdot w, a_1\cdot w\cdot d, x\cdot w') \in \eta_{\mathsf{C}}$, (5) $(x\cdot w', a_2\cdot w\cdot d, z\cdot w') \in \eta_{\mathsf{C}}$, (6) $(z\cdot w', b_2\cdot w\cdot d, y\cdot w') \in \eta_{\mathsf{C}}$. (7) By applying the induction hypothesis to (6) and (7), we obtain

$$(x \cdot w', l \cdot w \cdot d, y \cdot w') \in \eta_{\mathsf{C}}.$$

(8) By (5) and (8), we obtain $(x, c_2, y) \in \eta_{\mathsf{B}\Rightarrow\mathsf{C}}$.

■

Finally, we have

$(c_1 \cdot w \cdot d) \otimes (c_2 \cdot w \cdot d) = (a_1 \cdot w \cdot d) \otimes (k \cdot w \cdot d) \otimes (b_1 \cdot w \cdot d) \otimes (l \cdot w \cdot d)$ $\sqsupseteq (a_1 \cdot w \cdot d) \otimes (a_2 \cdot w \cdot d) \otimes (b_1 \cdot w \cdot d) \otimes (b_2 \cdot w \cdot e) \sqsupseteq (a \cdot w \cdot d) \otimes (b \cdot w \cdot d)$.

◀

Proof of Proposition 18.

The only interesting case is that of a function type $\mathsf{A} = \mathsf{B} \Rightarrow \mathsf{C}$. By Proposition 8, $(f, a, f') \in \rho^r_{\mathsf{A}}$ holds iff for all $a \sqsubseteq \widehat{q}^r_{\mathsf{A}}(f, f') = \widehat{q}^r_{\mathsf{A}}(f, f) \multimap \widehat{q}^r_{\mathsf{A}}(f, f')$, which is in turn equivalent to $a \otimes \widehat{q}^r_{\mathsf{A}}(f, f) \sqsubseteq \widehat{q}^r_{\mathsf{A}}(f, f')$. This implies then that $(f, a, f') \in \rho^r_{\mathsf{A}}$ iff for all $(f, b, f) \in \rho^r_{\mathsf{A}}$ (i.e. for all $b \sqsubseteq \widehat{q}^r_{\mathsf{A}}(f, f)$), $(f, a \otimes b, f') \in \rho^r_{\mathsf{A}}$ (i.e. $a \otimes b \sqsubseteq \widehat{q}^r_{\mathsf{A}}(f, f')$), that is, iff $(f, a, f') \in \delta_{\mathsf{A}}$. ◀

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